Submission LINALG1-MID-SUB01

Question LINALG1-MID-Q01
scratch: show step
1. det=2(3)–4(2)=-2
2. Eliminate y, yielding x=1; substitute ⇒ y=2
3. checks: 2(1)+4(2)=10, 2(1)+3(2)=8
=> FINAL: x=1, y=2 (check units/interpretation)

Question LINALG1-MID-Q02
scratch: show step
1. det(A)=6·4–2·2=20
2. det(A) ≠ 0, so A is invertible.
=> FINAL: det=20; invertible (check units/interpretation)

Question LINALG1-MID-Q03
scratch: show step
1. T(1,–1)=(1–2,3+1)=(–1,4)
2. Matrix [[1,2],[3,–1]]
3. Both coordinate rules have no constant term, so T is linear.
=> FINAL: T(1,-1)=(-1,4); matrix [[1,2],[3,-1]]; linear (check units/interpretation)

Question LINALG1-MID-Q04
scratch: show step
1. Set c■(1,0)+c■(0,1)=(0,0).
2. Coordinates give c■=0 and c■=0 only, so the set is independent.
=> FINAL: only trivial combination; {(1,0),(0,1)} is linearly independent (check units/interpretation)

Question LINALG1-MID-Q05
scratch: show step
1. Null(A) is the set of vectors sent to the zero vector.
2. It is the solution space of Ax=0 and is a subspace.
=> FINAL: the vectors mapped to zero; exactly the solution space of Ax=0 (check units/interpretation)

Question LINALG1-MID-Q06
scratch: show step
1. det=5(5)–2(2)=21
2. Eliminate y, yielding x=1; substitute ⇒ y=2
3. checks: 5(1)+2(2)=9, 2(1)+5(2)=12
=> FINAL: x=1, y=2 (check units/interpretation)

Question LINALG1-MID-Q07
scratch: show step
1. det(A)=2·2–2·2=0

2. det(A) = 0, so A is not invertible.
=> FINAL: det=0; not invertible (check units/interpretation)

Question LINALG1-MID-Q08
scratch: show step
1. T(1,–1)=(1–2,3+1)=(–1,4)
2. Matrix [[1,2],[3,–1]]
3. Both coordinate rules have no constant term, so T is linear.
=> FINAL: T(1,-1)=(-1,4); matrix [[1,2],[3,-1]]; linear (check units/interpretation)